Internetworking Devices

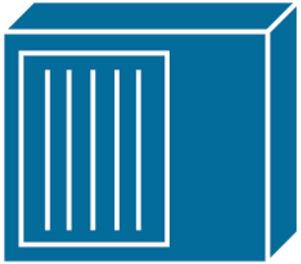
https://www.youtube.com/@techwithravish

Internetworking devices are hardware components used to connect different networks and manage data traffic between them. These devices operate at various layers of the OSI (Open Systems Interconnection) model and perform functions such as routing, switching, and bridging to ensure efficient data communication. Here are some common types of internetworking devices:

1. HUB
2. BRIDGE
3. Switch
4. Router
5. Wireless Access Point (WAP)

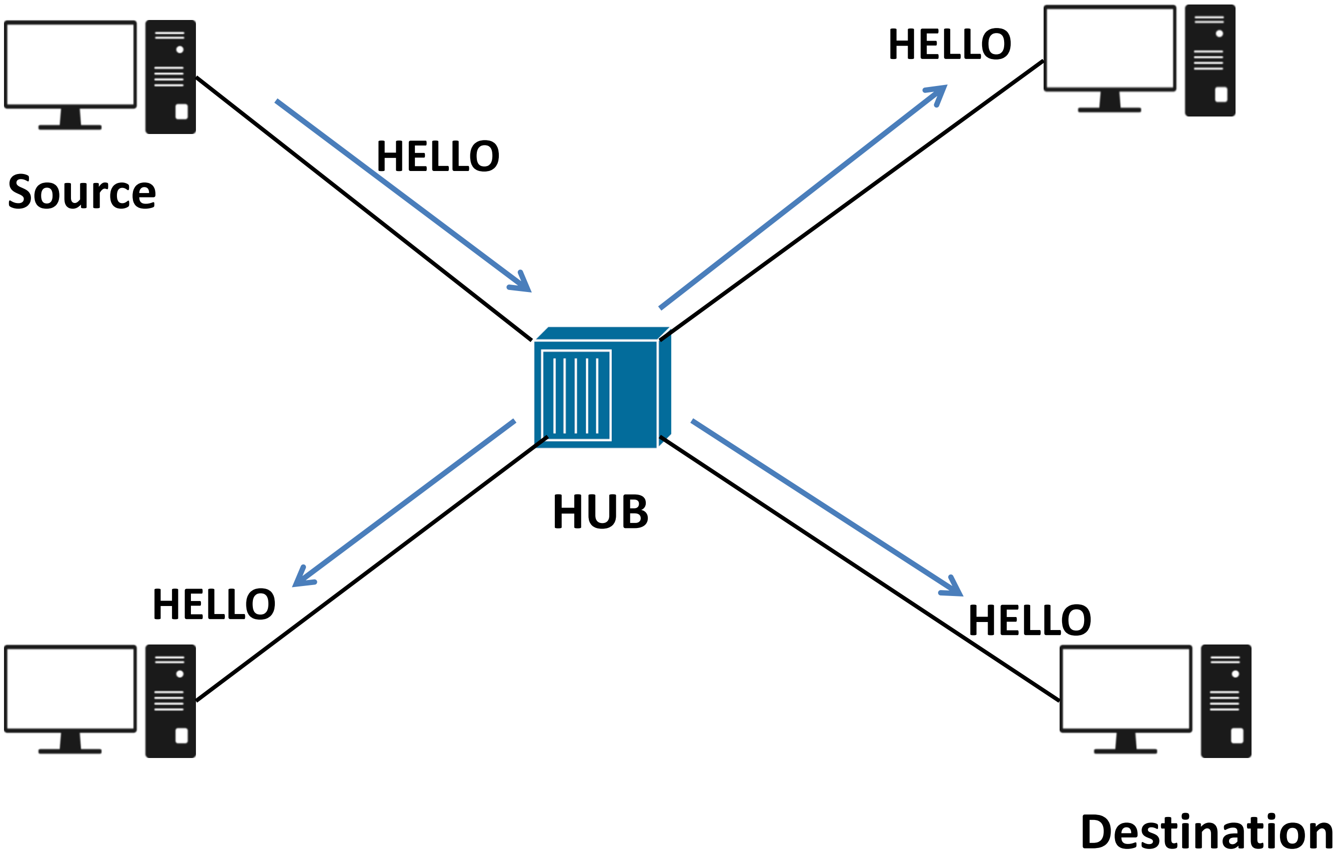
HUB:

-It is a Layer 1 Device
-Works on Physical layer
- Data Forwarded in Bits
- Plug and Play Device
- No. Of Ports: 2,4,6
- Half Duplex Device
- Operates in Single Broadcast Domain
- Always do Broadcast
- All ports collision



HUB

HUB Always do Broadcast:



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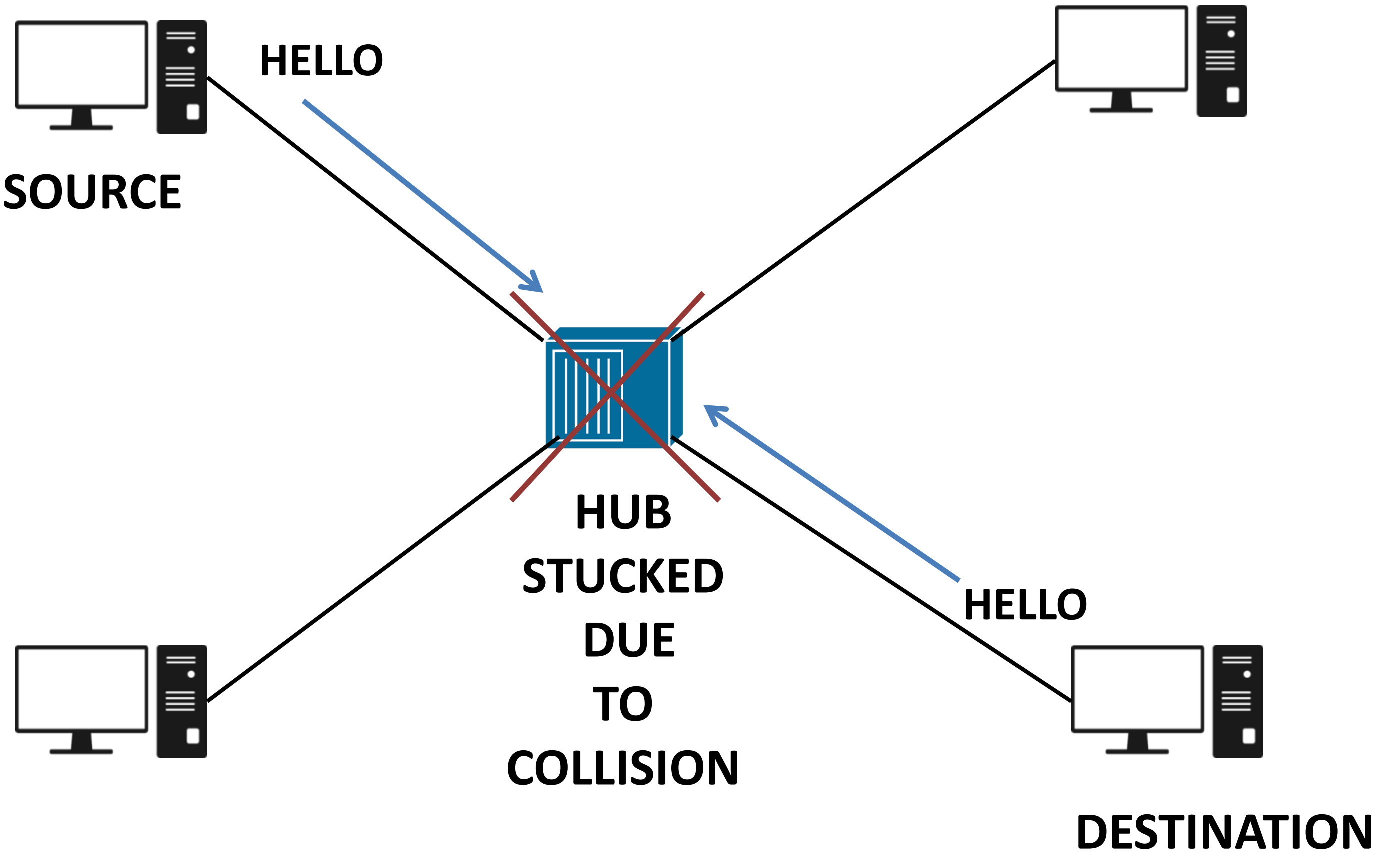
Collision in HUB: Only One PC can forwards data at one time



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Bridge:

- It is a Layer 2 Device
- Works on Data Link Layer
- Send or Receive Data Frame
- Learn MAC Address
- Plug and Play Device
- Single Broadcast Domain
- Per Port Collision
- No. Of Ports: 2
- Full Duplex Device

Bridge

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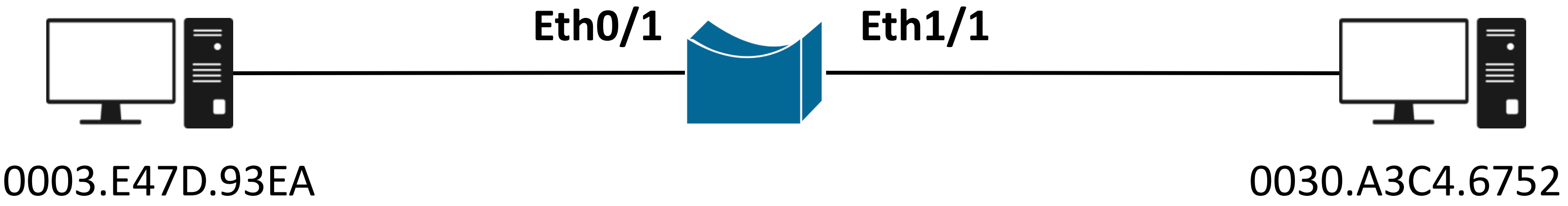
Bridge Maintains the MAC Address Table:



Port No.	MAC Address
Eth0/1	0003.E47D.93EA
Eth1/1	0030.A3C4.6752

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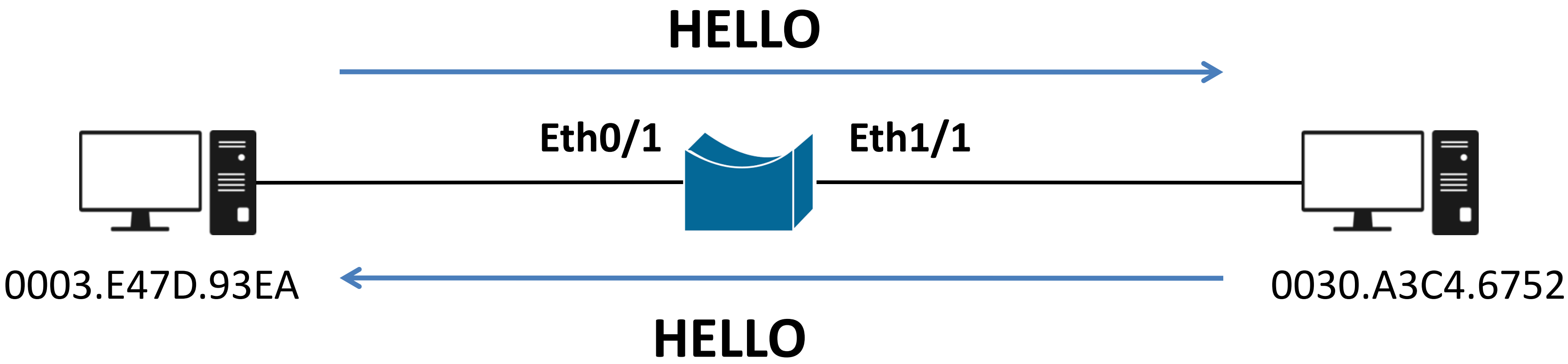
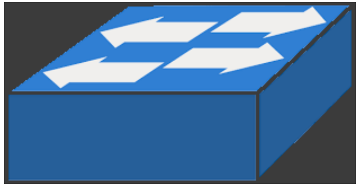
Bridge is a Full Duplex Device:



Switch: Two Types

- Layer 2 Switch and Layer 3 Switch
- Works on Data Link Layer
- Send or Receive Data Frame
- MAC Address Ageing Time 300 Sec
- Maintain MAC Address Table
- Switch has CLI and GUI Mode
- Single Broadcast Domain
- Per Port Collision
- No. Of Ports: 16, 24, 32, 48
- Full Duplex Device



Switch

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Cable and Port Type in Switch:

There are 4 Types of Link in Switch:

Link Type	Speed	Short Form	Command
Ethernet	10 MBPS	Eth	Int Eth0/0
Fast Ethernet	100 MBPS	Fa	Int Fa0/0
Gigabyte	1 GBPS	Gig	Int Gig0/0/0
Serial	1.54 MBPS	Se	Int Se0/1

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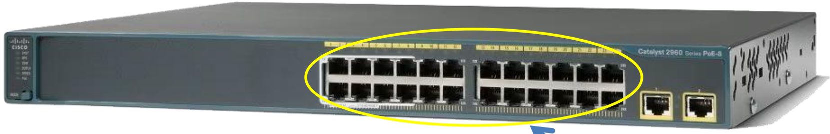
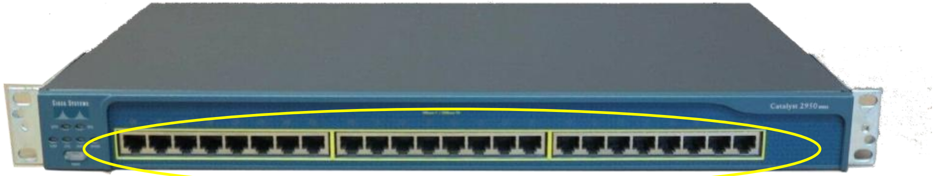
Cable and Port Type in Switch:

There are 4 Types of Link in Switch:



Catalyst 2960

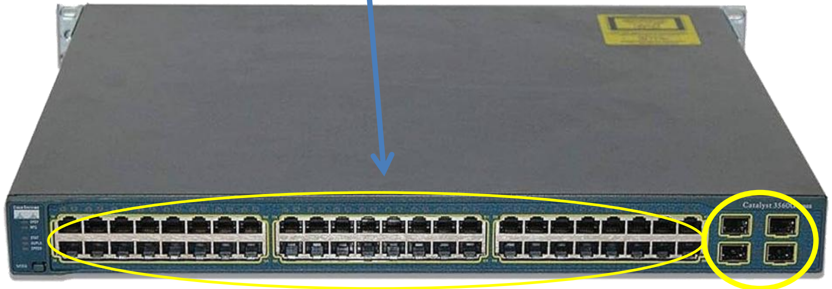


Catalyst 2950

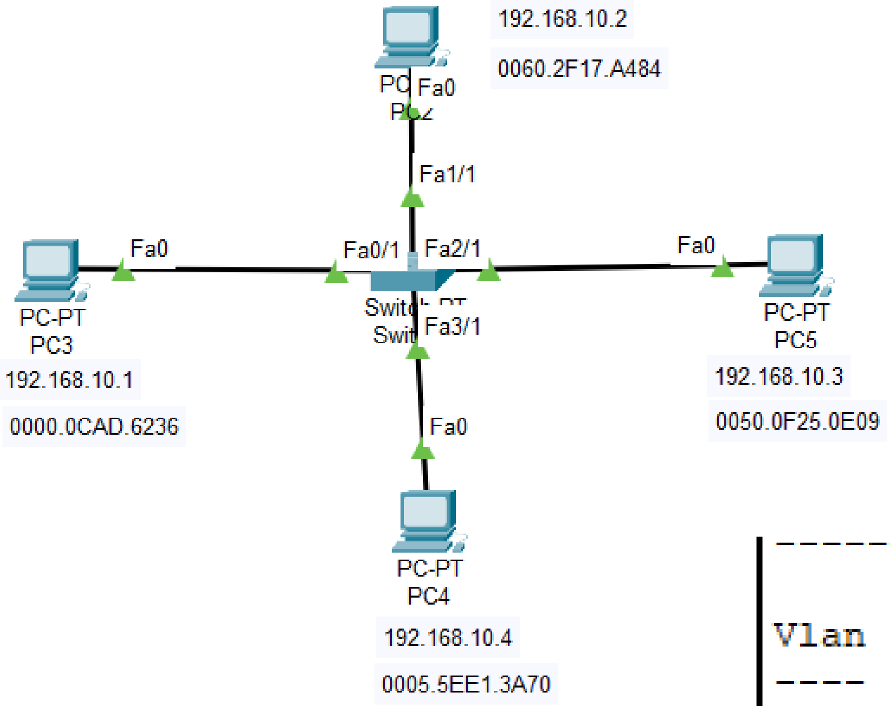
Fast Ethernet Port

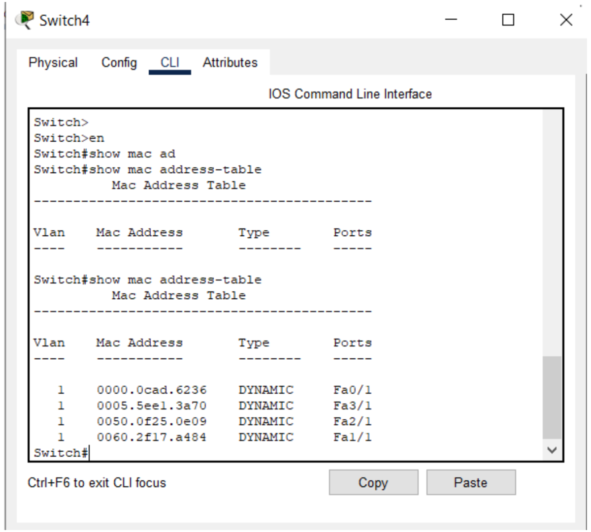
SFP Module

POE + SFP



Catalyst 3560



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Vlan    Mac Address       Type        Ports
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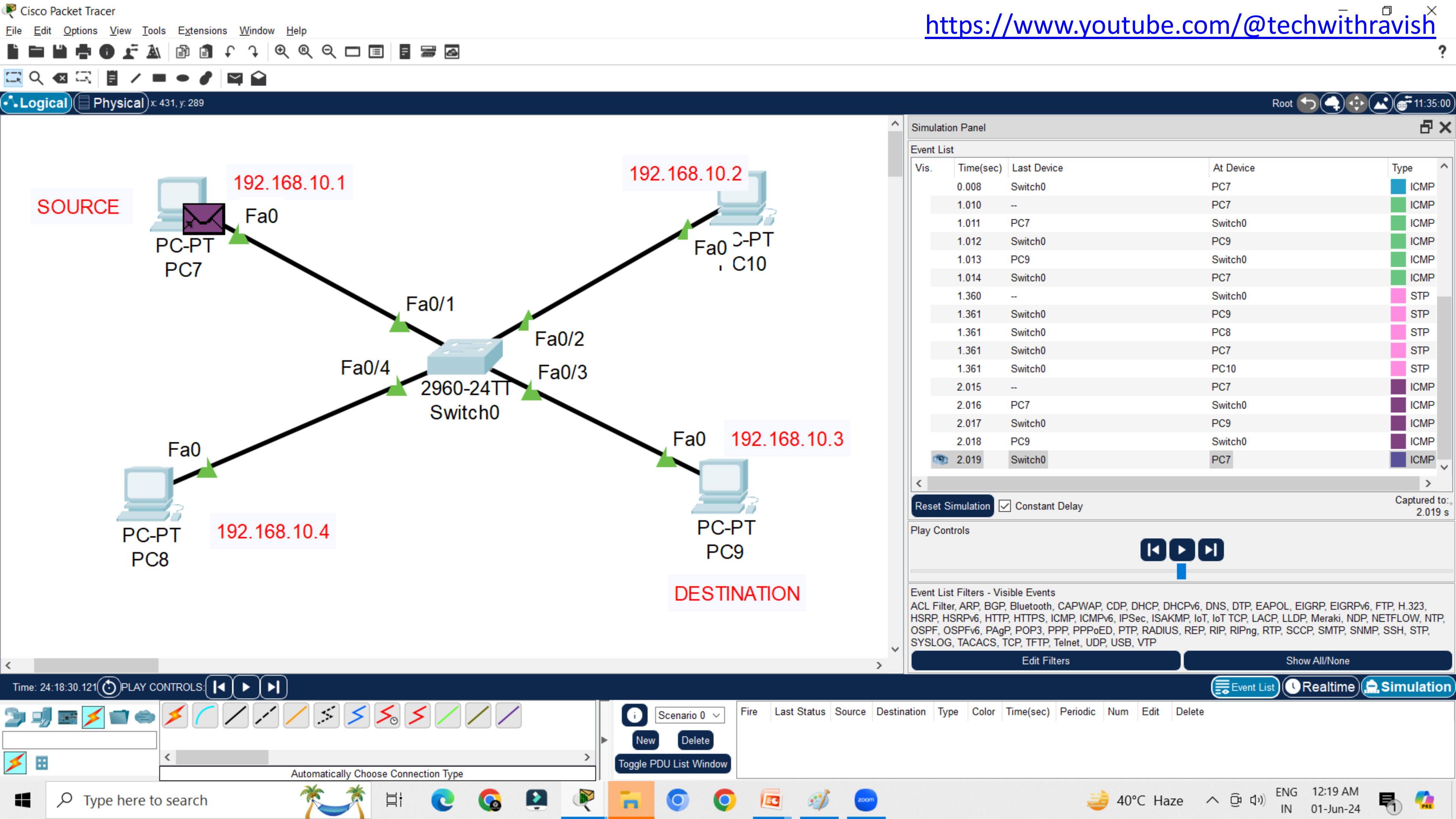
   1    0000.0cad.6236    DYNAMIC     Fa0/1
   1    0005.5ee1.3a70    DYNAMIC     Fa3/1
   1    0050.0f25.0e09    DYNAMIC     Fa2/1
   1    0060.2f17.a484    DYNAMIC     Fa1/1
Switch#
```

Command for MAC Address Table

Switch>en
Switch#show mac address-table



Cisco Packet Tracer
https://www.youtube.com/@techwithravish
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 431, y: 289
Root
11:35:00
SOURCE
192.168.10.1
Fa0
PC-PT
PC7
192.168.10.2
Fa0
C10
Fa0/1
Fa0/2
Fa0/4
Fa0/3
2960-24TT
Switch0
Fa0
192.168.10.3
Fa0
PC-PT
PC8
192.168.10.4
PC-PT
PC9
DESTINATION
Simulation Panel
Event List
Vis. Time(sec) Last Device At Device Type
0.008 Switch0 PC7 ICMP
1.010 -- PC7 ICMP
1.011 PC7 Switch0 ICMP
1.012 Switch0 PC9 ICMP
1.013 PC9 Switch0 ICMP
1.014 Switch0 PC7 ICMP
1.360 -- Switch0 STP
1.361 Switch0 PC9 STP
1.361 Switch0 PC8 STP
1.361 Switch0 PC7 STP
1.361 Switch0 PC10 STP
2.015 -- PC7 ICMP
2.016 PC7 Switch0 ICMP
2.017 Switch0 PC9 ICMP
2.018 PC9 Switch0 ICMP
2.019 Switch0 PC7 ICMP
Reset Simulation
Constant Delay
Captured to: 2.019 s
Play Controls
Event List Filters - Visible Events
ACL Filter, ARP, BGP, Bluetooth, CAPWAP, CDP, DHCP, DHCPv6, DNS, DTP, EAPOL, EIGRP, EIGRPv6, FTP, H.323, HSRP, HSRPv6, HTTP, HTTPS, ICMP, ICMPv6, IPSec, ISAKMP, IoT, IoT TCP, LACP, LLDP, Meraki, NDP, NETFLOW, NTP, OSPF, OSPFv6, PAgP, POP3, PPP, PPPoED, PTP, RADIUS, REP, RIP, RIPng, RTP, SCCP, SMTP, SNMP, SSH, STP, SYSLOG, TACACS, TCP, TFTP, Telnet, UDP, USB, VTP
Edit Filters
Show All/None
Time: 24:18:30.121
PLAY CONTROLS:
Event List
Realtime
Simulation
Scenario 0
New
Delete
Toggle PDU List Window
Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete
Automatically Choose Connection Type
Type here to search
40°C Haze
ENG
IN
12:19 AM
01-Jun-24